

The new pay Napster

Napster, the radical MP3 swapping application that started the peer-to-peer revolution, has opened a new chapter. Faced with dire financial situations, it reached an agreement with German media company Bertelsmann AG to offer a secure music file sharing service.



A beta program was started early this year, involving reworking of the software to integrate Digital Rights Management, accounts, audio fingerprinting, etc to work in tandem as part of the application. Despite a few spoilers like the download limit of 50 tracks a month and the inability to burn the downloaded music on CDs, the new Napster still feels like the old one. Its best feature—browsing through a real person's music collection, sending them messages and recommending new music—still survive. But the options to write songs on to a CD or move them to a portable player are noticeable by their absence, as is the option to search for MP3 files alone. The music player built into the software seems good enough, though the playlisting feature is a bit wobbly. But even with the ability to chat with other users and browse through their hard drives (albeit only the copy-right-cleared .nap tracks), it is tough to find tracks to download. But in terms of technology and how true it has stayed to the original model, the new Napster could still be a winner.

Blind games

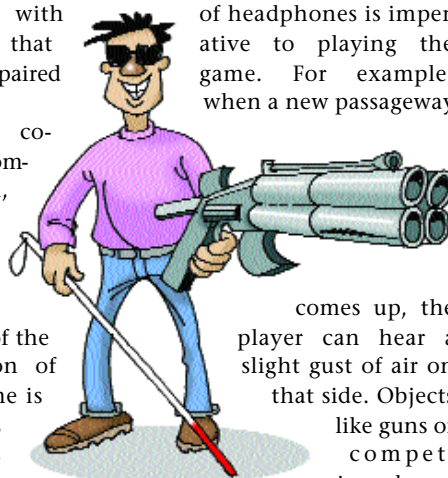
What started off as a friendship has developed into inspiration. Jeremie Spitzer and Paul Silva at the University of Massachusetts were hanging out with a visually impaired friend, Tim Keenan, when they decided to make computer games with such strong audio that even the visually impaired could play them.

The trio have co-founded a small company called ZForm, which is presenting its first product, an online video poker game, at the convention of the National Federation of the Blind. The game is quite simple: it tells the player the cards he is holding and what the other players are doing.

Spitzer maintains that the blind and sighted can com-

pete on the same level in this video game. ZForm is also building a prototype of *Quake*, the ultimate proof of their work.

Modifying *Quake* to create an environment with only sound cues hasn't exactly been child's play—a pair of headphones is imperative to playing the game. For example, when a new passageway



comes up, the player can hear a slight gust of air on that side. Objects like guns or competing players give off characteristic sounds, their direction indicated by their relative volume.

quoteworthy

"We're starting to shrink [chips] less rapidly than in the past"

Intel's Gordon Moore states that his prediction of chip performance doubling every two years may start to slow down from now on as the physical limits of processor fabrication are being reached

"Then we'll have options, and we like to have options"

Steve Jobs, on the fact that Apple is first concentrating on the company transitioning over to OS X before considering moving from using Motorola chips in their computers to Intel chips

snapshot

Losses due to software piracy in Asia-Pacific pegged at **\$4.7 billion** in 2001

Source: Gartner

tomorrow's technology

Autonomic computing

Researchers at IBM have identified a problem that has long been associated with computing—far too much time is spent in maintaining the system itself than getting the job at hand done. If computing systems can take care of themselves and adapt to situations as they arise, users would have more time to think about working with their actual applications. Inspiration for this computing philosophy was drawn from the most brilliant autonomous system

found in nature—the human body, which transparently takes care of all complex functioning like telling the heart when to beat, moderating temperature and repairing damages.

Autonomic computing will be a marriage of fields like artificial intelligence, neural networks, control theory as well as cybernetics. It promises systems with the ability to dynamically adjust to external conditions, diagnose errors and bottlenecks and

take appropriate action, protect itself from external threats, optimise and anticipate information delivery for the fastest possible operation—and all this is going to be transparent to users, enabling them to concentrate on the task at hand.

Working systems are already being demonstrated with 'cellular' chips that can recover from damage or failure and heterogeneous systems that can manage workloads over networks.